



SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences

Chennai-602105

A REPORT ON

Intelligent Water Distribution Monitoring and Leakage Detection Platform

SUBMITTED TO

SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

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Software Engineering

SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES



SUBMITTED BY

M Prasanth Kumar Reddy (192325053)

SUPERVISED BY

Dr. K. Anita Davamani



Saveetha

SIMATS ENGINEERING
Saveetha Institute of Medical and Technical Sciences
Chennai-602105



DECLARATION

I, M Prasanth Kumar Reddy, of the Department of Artificial Intelligence and Machine Learning, Saveetha Institute of Medical and Technical Sciences, Saveetha University, Chennai, hereby declare that the Report entitled 'Intelligent Water Distribution Monitoring and Leakage Detection Platform', submitted for the course CSA1012 - Software Engineering, is the result of my own bonafide efforts. To the best of my knowledge, the work presented herein is original, accurate, and has been carried out in accordance with the principles of engineering ethics.

Place: Chennai

Date: July 20, 2026

Signature of the Student with Name

M Prasanth Kumar Reddy

BONAFIDE CERTIFICATE

This is to certify that the report entitled **“Intelligent Water Distribution Monitoring and Leakage Detection Platform”** is a bonafide record of work carried out by **M Prasanth Kumar Reddy (Register No. 192325053)**, of the Department of Artificial Intelligence and Machine Learning, SIMATS Engineering, Saveetha Institute of Medical and Technical Sciences, Chennai, in partial fulfilment of the requirements for the course **CSA1012 - Software Engineering**, submitted as a Case Study Analysis Report on the assigned problem statement.

The report has been prepared under my supervision and, to the best of my knowledge, represents the student's own work and has not been submitted elsewhere for the award of any other credit.

Supervisor

Dr. K. Anita Davamani

Head of the Department

Department of AI & ML

Date: July 20, 2026

Place: Chennai

ABSTRACT

This report presents a systematic Case Study Analysis of the problem of undetected water loss in urban distribution networks, prepared in accordance with the Software Engineering case-study framework that requires problem understanding, stakeholder analysis, evaluation of the current approach, a proposed software-based solution, methodology justification, and a discussion of expected outcomes and limitations. Urban water utilities routinely lose treated water through pipeline leakages, aging infrastructure, and unauthorised consumption, and in most cities today a leak is confirmed only after a resident complains or physical damage becomes visible at the surface, by which point substantial water and revenue have already been lost.

The proposed solution is an Intelligent Water Distribution Monitoring and Leakage Detection Platform that continuously senses pipeline pressure, flow, and water-quality parameters using an IoT device layer, transmits this data securely to a cloud-based analytics engine, and applies predictive models to detect leaks, flag abnormal consumption, and forecast infrastructure failure before it occurs. The report explains how core software engineering principles, specifically modularity, scalability, maintainability, reliability, usability, and security, are incorporated into the design, and justifies the choice of an Agile development methodology for building and evolving the platform.

The analysis concludes with a discussion of the expected operational benefits of the platform, the risks and limitations that would need to be managed during implementation, and possible directions for future enhancement, giving a complete, end-to-end view of how a real-world infrastructure problem can be addressed through a disciplined software engineering approach.

Keywords: case study analysis, IoT water monitoring, leak detection, stakeholder analysis, software engineering principles, Agile methodology, predictive analytics.

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Section 1

Understanding the Problem Statement

1.1 The Problem, Explained in Simple Terms

Every city depends on a vast underground network of pipes to carry treated water from treatment plants to individual homes and businesses. Over years of use, these pipes develop small cracks and loose joints because of corrosion, pressure surges, ground movement, and nearby construction work. When a crack develops, water begins to leak out slowly and silently, usually well below the road surface where nobody can see it happening.

The core problem is that, today, most water utilities have no direct way of knowing that a leak has started. They typically find out about it in one of two ways: either a resident notices reduced water pressure or a wet patch on the road and calls to complain, or the leak grows large enough to burst through the surface and become physically visible. Both of these signals arrive late, often days or weeks after the leak first began, and by that time a large amount of treated water, along with the electricity and chemicals spent producing it, has already been wasted. In addition, because most of the pipeline network is old, utilities also struggle to know which specific sections of pipe are most likely to fail next, so repair and replacement work tends to happen reactively, after something has already gone wrong, rather than being planned in advance.

In short, the problem is a lack of early, continuous visibility into the physical health of the water network. This case study proposes to solve that problem through an intelligent, sensor-driven software platform, described in detail in Section 4 of this report.

1.2 Objectives of the Proposed System

The objectives below describe what the proposed system must achieve, and each one is explained in terms of the practical outcome it is meant to produce for the water utility.

- **Continuous monitoring:** the system must watch pipeline pressure, flow, and water quality around the clock, rather than relying on periodic manual inspection rounds, so that a problem is visible the moment it begins rather than days later.
- **Early leak detection:** the system must recognise the pressure and flow signature of a leak while it is still small, so that repair crews can be dispatched before the leak grows large enough to cause visible surface damage or major water loss.
- **Water quality monitoring:** the system must track turbidity and residual chlorine levels, because a compromised pipe wall can allow contaminants to enter the network, which is a public health concern as well as an infrastructure one.
- **Predictive failure forecasting:** the system must use historical repair records and sensor trends to estimate which pipeline segments are at the highest risk of failing soon, so that replacement budgets can be spent where they matter most instead of on a fixed age schedule.
- **Immediate maintenance notification:** once an issue is detected, the system must alert the correct maintenance team automatically, with enough location and severity detail that they can respond quickly rather than waiting for a manual report to be filed and forwarded.
- **Consumption pattern analysis:** the system must learn the normal usage pattern for each zone so that it can flag unusual spikes that may indicate unauthorised tapping or a fixture failure on the customer's side of the connection.
- **Operational reporting:** the system must produce dashboards and reports that let managers see, at a glance, how much water is being lost, where, and how quickly issues are being resolved.

1.3 Scope of the Problem

The scope of this case study covers the design of a city-wide smart water monitoring platform, from the physical sensors installed on the pipeline network through to the software that processes their data and the maintenance workflow that acts on the results. It includes the sensing layer (pressure sensors, flow meters, water quality sensors, and remotely operable smart valves), the communication layer that carries their data securely to the cloud, the analytics layer that turns raw readings into leak and failure predictions, and the application layer that utility staff use to monitor the network and respond to alerts. The scope is deliberately citywide rather than limited to a single pipeline, because the value of the platform comes specifically from its ability to give the utility a single, consistent view of the entire network at once, which also means the design must account for fault tolerance and high availability so that the failure of one device or link does not blind the system to an entire part of the city.

Section 2

Key Stakeholders and Challenges

A water distribution monitoring platform touches many different groups of people, each with a different relationship to the system and a different set of needs. Understanding these stakeholders clearly is essential before designing a solution, because a feature that helps one group may be irrelevant, or even burdensome, to another.

Stakeholder	Role and Responsibility in the System
Water Utility Management	Owns the overall outcome of the platform; responsible for budgeting the deployment, setting service-level targets
Control Room / Operations Staff	Uses the live dashboard on a daily basis to watch pressure and flow trends across the city, acknowledge alerts, and
Field Maintenance Crews	Receive automatically generated alerts on their mobile devices specifying the location and likely cause of an issue
Residents and Businesses (Consumers)	Experience the direct impact of the system through more reliable water pressure, fewer unexpected supply interruptions
IT and Platform Administrators	Responsible for keeping the sensors, gateways, and cloud services running, applying security updates, and ensuring
Software Development Team	Designs, builds, tests, and maintains the platform itself, including the sensor-integration code, the predictive model
Regulatory / Government Bodies	Set water-quality and reporting standards that the platform must help the utility comply with, and may also use ag

2.1 Major Challenges Faced by Each Stakeholder Group

Each stakeholder group faces a distinct set of difficulties under the current, largely manual way of managing the water network. Explaining these challenges in detail makes clear why a software-based solution is needed, and what it specifically needs to solve for each group.

- **Utility management** struggles to justify infrastructure investment because they lack reliable, granular data on exactly where and how water is being lost; without this evidence, it is difficult to prioritise which parts of the network most urgently need repair or replacement.
- **Control room staff** currently have very limited visibility into the real-time condition of the network; they typically only learn about a problem once a complaint call comes in, which means they are always reacting to events rather than anticipating them.
- **Field maintenance crews** often receive incomplete information about where a reported leak actually is, leading to wasted travel time searching for the source of a problem that could have been pinpointed automatically by sensor data.
- **Residents** experience unpredictable low-pressure periods or supply interruptions with no advance warning, and may also unfairly bear the cost of network losses through higher water tariffs, since non-revenue water is ultimately paid for by all customers.
- **IT and platform administrators** face the challenge of keeping a very large number of physically distributed, low-power devices online and secure, particularly given that some of these devices can remotely operate valves, which raises the stakes of any security lapse.
- **The development team** must build a system that works reliably across a huge range of field conditions, from dense urban districts with strong connectivity to remote outskirts where network coverage is poor, without compromising on data accuracy or response time.

Section 3

Analysis of the Current Approach

3.1 How Water Networks Are Monitored Today

Most water utilities currently rely on a combination of three practices to keep track of the condition of their distribution network. First, periodic manual inspection rounds are carried out along accessible sections of pipeline, where a technician visually checks for surface signs of a leak, such as damp patches or unusually lush vegetation above a pipe. Second, a small number of fixed telemetry points, typically at major trunk mains, feed pressure and flow readings back to a central SCADA (Supervisory Control and Data Acquisition) system on a set schedule rather than continuously. Third, and most heavily relied upon in practice, the utility waits for customer complaints, whether about low pressure, a visible leak, or an unusually high bill, and dispatches a crew to investigate only after such a complaint is logged.

Between these inspection rounds and telemetry checkpoints, the vast majority of the pipeline network, especially smaller distribution mains that reach individual streets and buildings, remains effectively unmonitored. A leak in one of these unmonitored sections can therefore persist for a long time before anyone becomes aware of it.

3.2 Strengths of the Current Approach

The existing approach is not without merit, and it is important to recognise what it does well before proposing a replacement. Manual inspection benefits from human judgement and can catch problems, such as visible corrosion on an exposed fitting, that a sensor might not be configured to detect. SCADA systems at trunk mains are mature, well-understood technology with a long track record of reliable operation for the specific high-value assets they cover. Relying on customer complaints, while slow, is also essentially free in terms of monitoring infrastructure cost, since it uses the affected residents themselves as an ad-hoc sensor network.

3.3 Limitations and Gaps

- **Sparse coverage:** fixed telemetry exists only at a small number of trunk-main locations, leaving the much larger network of smaller distribution pipes essentially invisible to the utility between manual inspection rounds.
- **Slow detection:** because the primary detection mechanism is a customer complaint, there is an inherent delay between when a leak starts and when the utility becomes aware of it, during which water continues to be lost.
- **No predictive capability:** the current approach is almost entirely reactive; there is no systematic way to identify which pipeline segments are approaching failure before they actually fail, so maintenance budgets are spent responding to emergencies rather than preventing them.
- **Manual, labour-intensive processes:** inspection rounds and complaint handling both require significant staff time, which limits how frequently any given section of the network can realistically be checked.
- **Fragmented data:** readings from SCADA checkpoints, inspection logs, and complaint records are typically stored in separate systems, making it difficult to build a single, coherent picture of network health over time.

These gaps directly motivate the software solution proposed in Section 4: a platform that extends continuous monitoring across the entire network, shortens detection time from days to minutes, and adds a predictive capability that the current approach lacks entirely.

Section 4

Proposed Software Solution

The proposed solution is the Intelligent Water Distribution Monitoring and Leakage Detection Platform, a layered software system that continuously ingests data from IoT devices installed across the pipeline network, applies predictive analytics to that data, and automatically notifies maintenance teams when action is needed. The platform is organised into distinct functional modules so that each concern can be built, tested, and scaled independently.

4.1 Major Functional Modules

Module	What It Does and Why It Matters
Sensor Management	Keeps track of every device deployed in the field, its location and calibration status, and raises a device-health alert if a sensor goes offline.
Data Ingestion	Receives incoming readings from edge gateways, checks that each reading falls within a plausible range, and publishes clean data to the analytics layer.
Leak Detection Engine	Continuously scores incoming pressure and flow data against a trained model of normal behaviour and raises a leak event if the score exceeds a threshold.
Consumption Analysis	Builds a rolling profile of expected usage for each zone and flags statistically significant deviations, which may indicate a leak or illegal tapping.
Predictive Maintenance	Uses pipe age, material, and historical repair data to score each pipeline segment's risk of failure, giving planning staff a prioritised list of areas to inspect.
Alert and Notification	Converts a detection into a structured alert, works out which maintenance team is responsible for that zone, and sends the alert via email, SMS, or a mobile app.
Operations Dashboard	Gives control-room staff a live map, trend charts, and an alert queue in one place, and lets authorised operators issue a remote valve closure if needed.

4.2 Functional and Non-Functional Requirements

Functional requirements describe what the system must do, while non-functional requirements describe how well it must do it. Both are listed below with a short explanation of why each one is necessary for the platform to be genuinely useful to a utility.

Functional Requirements

- The system shall collect pressure, flow, and water-quality readings from all connected sensors at a configurable interval, so that the analytics engine always has current data to work with.
- The system shall detect leakage events and generate an alert with the estimated location and severity, so that maintenance staff know both that a problem exists and roughly where to look.
- The system shall allow an authorised operator to remotely close a smart valve, so that a confirmed leak segment can be isolated quickly while repair is arranged.
- The system shall provide a dashboard showing live and historical data for any selected zone, so that both day-to-day operators and long-term planners can use the same underlying data.
- The system shall generate scheduled operational reports summarising water loss and response times, so that management has the evidence needed for infrastructure investment decisions.

Non-Functional Requirements

- **Reliability:** the system must continue detecting leaks correctly even when individual sensors or network links fail, since a monitoring platform that goes blind during a fault is of little value.
- **Scalability:** the system must support growth from a pilot deployment of a few hundred sensors to a full city-wide rollout of tens of thousands, without requiring a redesign.
- **Security:** all communication, especially commands sent to smart valves, must be encrypted and authenticated, because an insecure control channel on physical infrastructure is a serious safety risk.
- **Usability:** the dashboard must be understandable to control-room staff without extensive technical training, since operational decisions often need to be made quickly under pressure.
- **Performance:** the time between a leak occurring and an alert being raised should be within a few minutes, since the entire value of the platform depends on detecting problems early.

4.3 How Software Engineering Principles Are Applied

Good software engineering practice is not just about making the system work once; it is about making sure it keeps working correctly as it grows, changes, and faces failures. The table below explains, principle by principle, exactly how the design choices described earlier put each principle into practice.

Principle	How It Is Applied in This Platform
Modularity	The system is split into independent modules, such as sensor management, leak detection, and notification, each with a clear interface.
Scalability	Sensors publish to a message broker that can absorb bursts of data, and the cloud services are containerised so that additional instances can be added easily.
Maintainability	Clear module boundaries, consistent logging, and a documented data schema mean that a developer joining the project later can understand and modify the system.
Reliability	Edge gateways buffer readings locally during a network outage and resynchronise automatically afterward, and cloud services are replicated for high availability.
Usability	The operations dashboard is designed around the tasks control-room staff actually perform, live map, alert queue, and trend analysis.
Security	Every device authenticates with its own certificate rather than a shared password, all data in transit is encrypted, and role-based access control is enforced.

Section 5

Justification of the Chosen Methodology

This project recommends the **Agile methodology**, delivered through short, iterative sprints, as the most suitable software development approach for building and evolving this platform, in preference to a traditional Waterfall model.

Why Agile Fits This Project

- **The requirements will evolve as the pilot runs:** exactly which pressure-drop pattern reliably indicates a leak, for example, is something that can only be tuned once real sensor data starts coming in from a pilot zone, which favours a methodology built around frequent feedback and revision rather than a fixed, upfront specification.
- **Independent modules suit independent sprints:** because the platform is already broken into modules such as sensor management, leak detection, and notification, as described in Section 4.1, these modules map naturally onto separate sprint backlogs that different parts of the development team can work on in parallel and integrate incrementally.
- **Stakeholders need to see working software early:** utility management and control-room staff, identified in Section 2, will find it far easier to give useful feedback on a working dashboard prototype after a few weeks than on a written specification document, and Agile's short release cycles make that early feedback loop possible.
- **Risk is reduced through incremental delivery:** deploying a small pilot zone first, gathering results, and then expanding city-wide, matches the natural rhythm of Agile sprints and lets problems be caught and corrected while the cost of change is still low, rather than discovering an issue only after a full city-wide rollout has already been built.

By contrast, a Waterfall approach, which requires the full set of requirements to be fixed before development begins, would be a poor fit here: the exact detection thresholds and dashboard workflows genuinely cannot be known with confidence until real operational data and user feedback are available, and Waterfall offers no efficient way to revise earlier decisions once later stages are underway.

Section 6

Expected Outcomes and Limitations

6.1 Expected Benefits and How They Address the Original Challenges

Each expected benefit below is linked back to the specific challenge it is designed to resolve, so that the connection between the proposed solution and the original problem is explicit rather than assumed.

- **Reduced non-revenue water losses:** because leaks are detected within minutes rather than days, directly answering the slow-detection limitation identified in Section 3.3, far less treated water escapes before a repair crew is dispatched.
- **Lower emergency repair costs:** catching a leak while it is still small means repairs are simpler and cheaper than the excavation and road repair often required once a leak has grown large enough to surface on its own.
- **Better-targeted infrastructure investment:** the predictive maintenance module directly solves the “no predictive capability” gap noted earlier, letting utility management, identified in Section 2, prioritise replacement budgets using evidence rather than guesswork.
- **Improved service reliability for residents:** fewer undetected leaks and faster repair response mean fewer unexpected pressure drops and supply interruptions for the consumer stakeholder group.
- **Less manual inspection workload:** continuous automated monitoring reduces the utility's reliance on labour-intensive manual inspection rounds, freeing staff time for higher-value work.

6.2 Risks, Implementation Challenges and Limitations

- **Sensor deployment cost:** installing pressure, flow, and quality sensors across an entire city is a significant upfront capital expense, which means most utilities would realistically need to roll the platform out zone by zone rather than everywhere at once.
- **Model accuracy during the early pilot period:** the leak-detection model needs real operational data to reach high accuracy, so during the first months of deployment it may produce some false alarms or miss subtle leaks until it has learned the specific pressure behaviour of the local network.
- **Connectivity gaps in remote areas:** some outlying parts of a city may have poor cellular or network coverage, which can delay data delivery from those zones even with local buffering at the edge gateway.
- **Security of remotely operable valves:** because the platform can physically act on the network by closing a valve, any weakness in the command channel's security is a more serious concern than in a purely observational system, and must be addressed carefully.

- **Organisational change management:** control-room staff and field crews who are used to a complaint-driven workflow will need training and time to adjust to an alert-driven one, and resistance to that change is a realistic implementation risk.

6.3 Future Enhancements

Once the core platform is operating reliably, it could be extended with satellite or drone-based thermal imagery to corroborate ground-sensor detections over very large service areas, with a digital-twin simulation of the network to test the effect of proposed interventions before they are carried out physically, and with a citizen-facing mobile application that lets residents report a suspected leak directly into the same alert pipeline used by the platform itself.

6.4 System Architecture Diagram

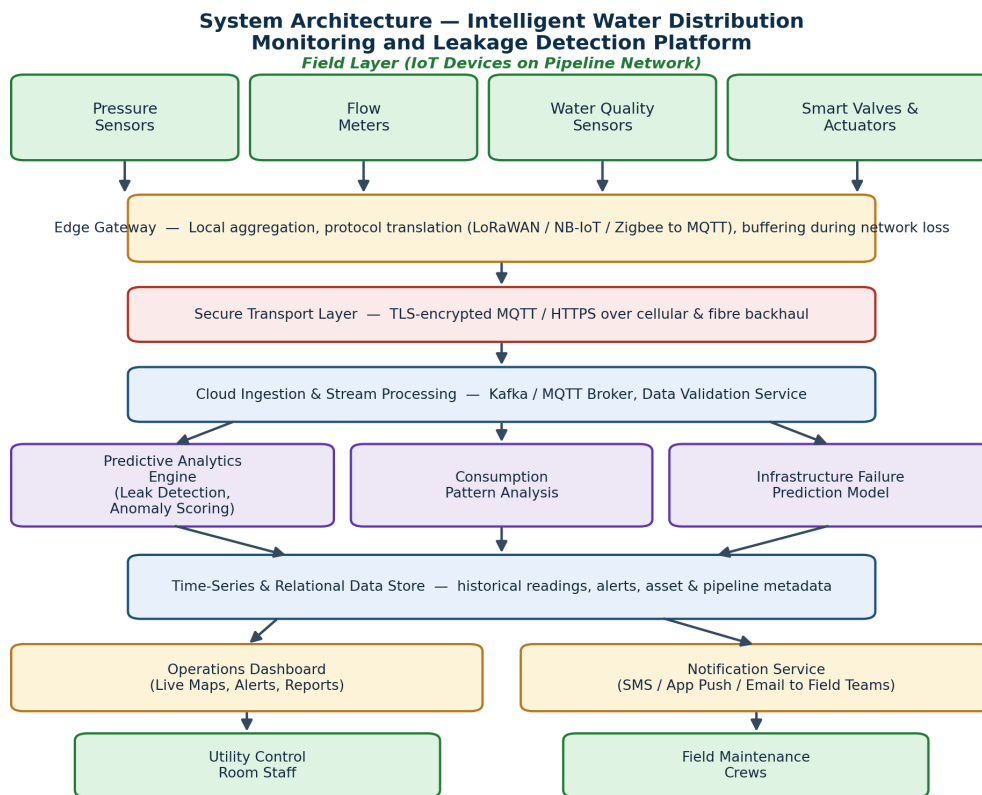


Figure 6.1: End-to-end architecture, from field-level IoT sensors and smart valves through the edge gateway, secure transport, cloud ingestion and analytics layers, to the data store, operations dashboard, and the utility staff who consume the platform's output.

6.5 Illustrative Operations Dashboard

The mockup below illustrates what the operations dashboard described in Section 4.1 would show to control-room staff during an active leak event: a live pressure trend for the affected zone, key operational indicators, a zone-wise comparison of estimated water loss, and a map of currently active leak alerts across the monitored area.

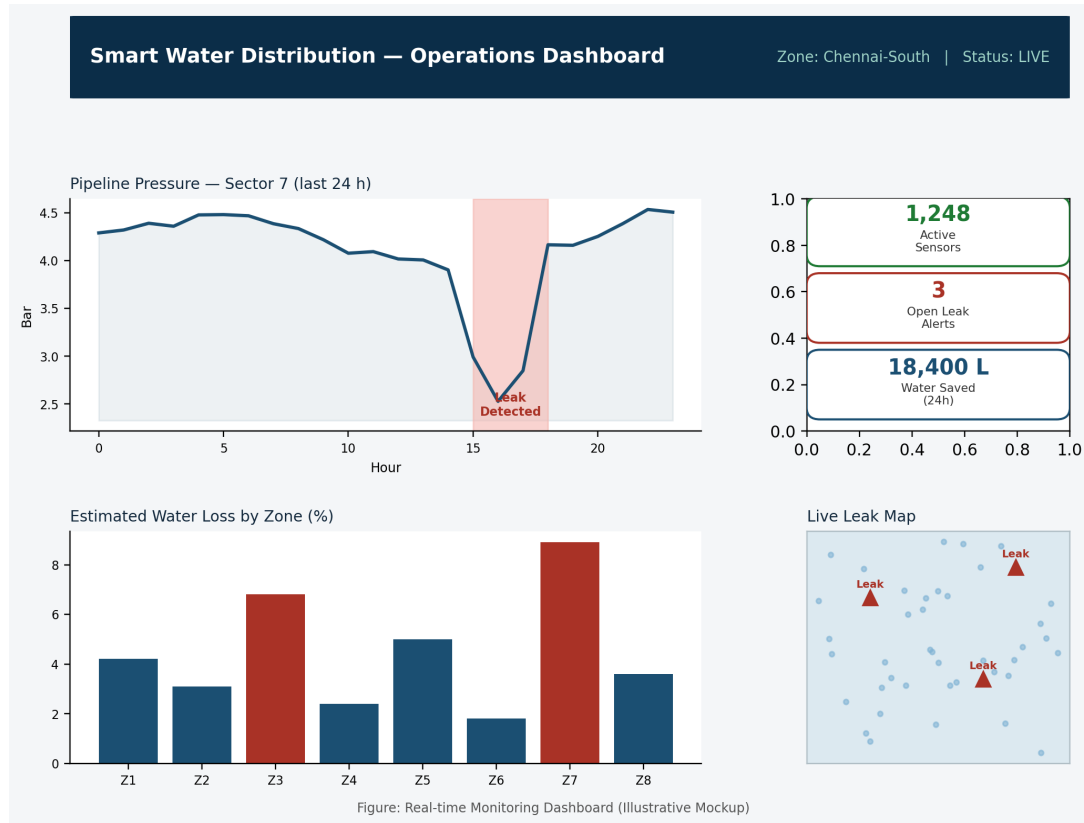


Figure 6.2: Illustrative real-time operations dashboard showing a live pressure trend with a detected leak event, key operational indicators, zone-wise estimated water loss, and a map of currently active leak alerts across the monitored area.

Section 7

Conclusion and References

This case study has worked through the assigned water-distribution problem systematically: explaining the problem and its scope in plain terms, identifying every stakeholder group and the specific challenges each one faces, analysing the strengths and gaps of the current, largely manual approach, and proposing a modular, secure, and scalable software platform built around continuous IoT sensing and predictive analytics. The Agile methodology was justified as the best fit for a project whose detection thresholds and user workflows can only be refined through real operational feedback, and the expected benefits of the proposed platform were traced directly back to the limitations identified in the current-system analysis, alongside an honest discussion of the risks and limitations that would need to be managed during a real implementation.

Overall, the analysis demonstrates that a disciplined software engineering approach, one that starts from a clear understanding of stakeholders and requirements rather than jumping straight to a technical design, can turn a longstanding infrastructure problem into a well-scoped, buildable software solution.

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